

Uncertainty-Robust Runtime Authorization for Grid AI Agent Actions: Worst-Case Verification via Interval Power Flow

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Abstract

Abstract: Runtime authorization layers that verify AI agent actions in a network model before release are emerging as the safety mechanism of choice for LLM-assisted grid operation, but existing layers verify against a single nominal model and therefore inherit its measurement and parameter errors. We propose an uncertainty-robust authorization layer that verifies each proposed action against the exact worst case of a bounded load uncertainty box: because the DC power flow is linear in the injections, the worst case on every branch is solved exactly by linear programming, giving a constructive guarantee—every load realization inside the box is safe when the action is released—rather than a statistical approximation. The required network model is identified from telemetry alone (voltage-angle differences and active flows), reproducing AC branch flows with Pearson correlation 0.9997. On the `rte_case14_realistic` system, the interval layer releases zero actions that are dangerous under a $\pm 20\%$ measurement mismatch at a false-reject cost of 1.02% and 42.7 ms per action, whereas nominal deterministic-twin verification releases 17.96% of such actions and Monte Carlo verification 2.55%; the zero false-pass property holds across all thirteen ablation configurations, degrades monotonically when the assumed uncertainty underestimates the true mismatch, and transfers to a 36-bus system. An LLM proposer that freezes under self-verification prompting acts under verifier-aware prompting and is rendered safe by the layer.

Keywords: LLM agents; power grid; runtime verification; interval analysis; power flow; safety.

1 Introduction

Large language model (LLM) agents are moving from research prototypes to operational decision-support tools in power systems. Closed-loop frameworks such as SafeVolt [1] use LLM agents for voltage control in active distribution networks, and dispatcher-assistant systems are being piloted on real control-room workflows. A central obstacle, stated explicitly by these works, is that *syntactic validity does not imply physical admissibility*: an agent that produces well-formed grid commands can still propose actions that overload branches or island the network.

To address this, runtime authorization layers that verify each proposed action in a network model before release have been proposed, following a line of work on learning agents for grid operation that started with the L2RPN challenge [2]. TwinGridShield [3] checks connectivity, branch flows, and generator limits in a deterministic network twin and releases an action only if the nominal model declares it safe. SafeVolt [1] evaluates candidate actions in a nominal simulator and filters them with a fine-tuned judge. Benchmarks such as ElecBench [4] and PowerAgentBench [5,6] now score safety compliance explicitly. What none of these mechanisms provides, however, is a safety guarantee that survives the gap between the model used for verification and the physical system: measurement noise, load forecast errors, and drifted line ratings. When the verification model differs from the true system, a nominal check can certify an action as safe that is in fact dangerous. The verification layer then inherits the model error, and the safety property holds only for the encoded nominal scenario.

In this paper we close this gap with an uncertainty-robust authorization layer whose decision is derived from the worst case over a bounded uncertainty set of load injections. Our contributions are as follows.

- We formulate the authorization problem over an interval model of the grid: branch flows are verified against their exact worst case over the load uncertainty box. Because the DC power flow is linear in the injections, the worst case is an exact linear program (LP), not a statistical approximation; the resulting guarantee is constructive: every load realization inside the box is safe when the action is released.
- We show that the DC network model required by the verification layer can be identified from telemetry alone (voltage-angle differences and active branch flows), removing the need for an offline network file, and we measure its fidelity against AC power flow on a standard test system.
- We quantify the vulnerability of nominal verification under measurement mismatch, and the behavior of an LLM proposer under two prompting protocols: a self-verifying prompt, under which the LLM freezes (proposes no actions), and a verifier-aware prompt, under which it acts; the authorization layer makes the latter safe.
- We evaluate the trade-off between safety and conservatism (false-reject rate) across uncertainty widths, verification quantities, network-model sources, proposers, and safety margins, and we demonstrate the approach on three systems of different sizes. We further show that every rejection is accompanied by a checkable counterexample (the load pattern that attains the violation) and that the uncertainty width can be calibrated from measurement data with a finite-sample coverage bound.

The remainder of the paper is organized as follows. Section 2 presents the problem formulation, the telemetry-based identification, and the interval verification layer. Section 3 reports the main comparison, the LLM proposer study, ablations, robustness, and cross-system results. Section 4 discusses implications and limitations, and Section 5 concludes.

2 Methods

2.1 Problem Formulation

We model the grid by its DC approximation over the set of substations $\mathcal{N}$ (with slack bus s_0) and branches $\mathcal{L}$. Each branch $\ell \in \mathcal{L}$ connects substations $o(\ell)$ and $e(\ell)$ with reactance x_ℓ . Given nodal injections $\mathbf{P} \in \mathbb{R}^{|\mathcal{N}|}$, the DC power flow solves

$$\mathbf{B}\boldsymbol{\theta} = \mathbf{P}, \quad f_\ell = \frac{\theta_{o(\ell)} - \theta_{e(\ell)}}{x_\ell}, \quad (1)$$

where $\mathbf{B}$ is the susceptance matrix and $\boldsymbol{\theta}$ the vector of voltage angles (the slack angle is fixed to zero). A proposed action is a set of branch disconnections; it transforms the network by removing the corresponding branches (equivalently, $x_\ell \rightarrow \infty$), which we denote by $\mathcal{L}'$.

The safety predicate for the post-action network is

$$\max_{\ell \in \mathcal{L}'} |f_\ell| \leq (1 - m) \bar{f}_\ell, \quad (2)$$

where $\bar{f}_\ell$ is the thermal rating of branch ℓ and $m \in [0, 1)$ is a safety margin. An action that disconnects the network (creates islands) is rejected outright.

The key modeling choice is the representation of measurement uncertainty. We assume each load is observed up to a symmetric relative error ε , so that the true injections lie in the box

$$\mathcal{P} = \{\mathbf{P} : P_i \in [P_i^-, P_i^+], \forall i\}, \quad (3)$$

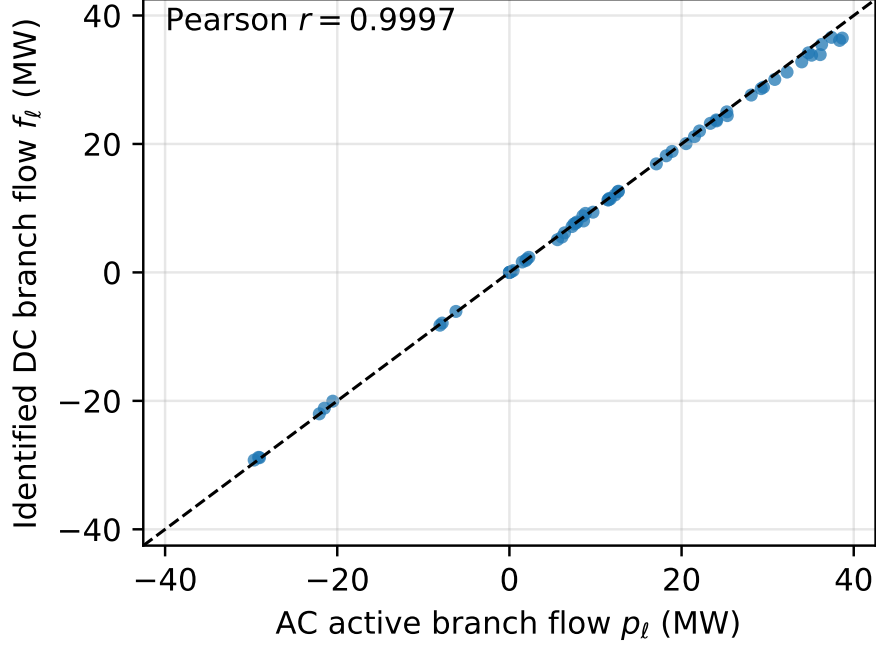


Figure 1: Identified DC branch flows versus AC branch flows on `rte_case14_realistic` (three states, 60 branch observations). The identified model reproduces AC flows with Pearson correlation $r = 0.9997$.

where $P_i^\pm = P_i^0 \pm \varepsilon |P_i^0|$ for load buses and $P_i^- = P_i^+ = P_i^0$ for generation buses. This box model follows the interval-analysis tradition [7], with the distinction that we solve for the exact image of the box rather than an enclosure. Because the mapping $\mathbf{P} \mapsto f_\ell$ in (1) is linear, the image of the box is an interval $[f_\ell^-, f_\ell^+]$ on each branch, and the exact endpoints are the solutions of two linear programs (LPs) per branch, solved with a simplex implementation [8]:

$$f_\ell^- = \min_{\mathbf{P} \in \mathcal{P}} f_\ell(\mathbf{P}), \quad f_\ell^+ = \max_{\mathbf{P} \in \mathcal{P}} f_\ell(\mathbf{P}). \quad (4)$$

2.2 Telemetry-Based Network Identification

The verification layer requires $\mathbf{B}$, i.e., the branch reactances and the substation topology. Instead of an offline network file, we identify the model from telemetry. Grid operating measurements routinely include branch voltage angles and active flows; on branch ℓ , the DC relation gives $x_\ell \approx (\theta_{o(\ell)} - \theta_{e(\ell)})/f_\ell$ per snapshot. We collect T snapshots and estimate each reactance by least squares,

$$\hat{x}_\ell = \frac{\sum_{t=1}^T \Delta \theta_{\ell,t} f_{\ell,t}}{\sum_{t=1}^T f_{\ell,t}^2}, \quad (5)$$

which is robust to snapshot noise. The topology (branch endpoints) is read from the same telemetry records.

2.3 Interval Verification and the Constructive Guarantee

Algorithm 1 summarizes the authorization layer.

Lemma 1 (Constructive safety). *If Algorithm 1 releases the action, then every load realization $\mathbf{P} \in \mathcal{P}$ produces a post-action power flow satisfying (2).*

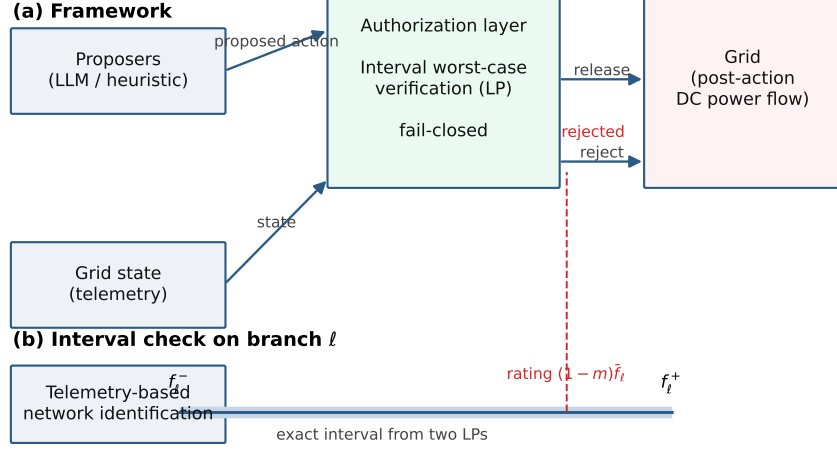


Figure 2: Framework: proposers submit actions, the authorization layer releases them only if the exact worst case of the post-action branch-flow interval over the load uncertainty box satisfies the rating constraint (fail-closed on any numerical failure). The network model is identified from telemetry.

Algorithm 1 Interval authorization of a proposed action

Require: Proposed disconnections $\mathcal{D}$; identified model $(\mathcal{L}, \mathbf{B}, \bar{\mathbf{f}})$; box $\mathcal{P}$; margin m .

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 $\mathcal{L}' \leftarrow \mathcal{L} \setminus \mathcal{D}$ 
if  $\mathcal{L}'$  disconnects  $\mathcal{N}$  then
    return reject
end if
for each branch  $\ell \in \mathcal{L}'$  do
     $f_\ell^- \leftarrow \min_{\mathbf{P} \in \mathcal{P}} f_\ell(\mathbf{P})$  (LP)
     $f_\ell^+ \leftarrow \max_{\mathbf{P} \in \mathcal{P}} f_\ell(\mathbf{P})$  (LP)
    if  $\max(|f_\ell^-|, |f_\ell^+|) > (1 - m) \bar{f}_\ell$  then
        return reject
    end if
end for
return release

```

100 *Proof.* By linearity of (1), for every $\mathbf{P} \in \mathcal{P}$ and branch ℓ we have $f_\ell(\mathbf{P}) \in [f_\ell^-, f_\ell^+]$, where the
 101 endpoints are exact (LP optima). The release condition checks $\max(|f_\ell^-|, |f_\ell^+|) \leq (1 - m) \bar{f}_\ell$ on
 102 every branch; hence (2) holds for every $\mathbf{P}$ in the box. $\blacksquare$ $\square$

103 The guarantee is constructive rather than statistical: it does not depend on the sampling
 104 distribution inside the box and cannot be invalidated by an unlucky sample. The price is
 105 conservatism: actions that are safe for most—but not all—loads in the box are rejected, which
 106 we measure as the false-reject rate.

107 A second consequence of linearity is that every rejection can be accompanied by evidence.
 108 For a rejected action, the LP optimizer that found the violating bound also returns the argmax
 109 load vector $\mathbf{P}^* \in \mathcal{P}$; by linearity, $\mathbf{P}^*$ attains the branch flow that violates (2). The authorization
 110 layer therefore reports not only that an action is rejected but *why*: “under load pattern $\mathbf{P}^*$,
 111 branch ℓ carries $f_\ell(\mathbf{P}^*) > (1 - m) \bar{f}_\ell$ ”. This witness is a concrete, checkable explanation of a
 112 safety decision—an operator or an auditing LLM can verify it by re-running a single power flow.
 113 The computational cost of the layer is $2|\mathcal{L}|$ LPs per action plus the witness extraction, which
 114 reuses the final LP solve of the violating branch.

2.4 Baseline Verifiers

- **No verification** (B_0): proposals are released unconditionally.
- **Nominal verification** (B_1): the action is checked against the single nominal injection point, i.e., (2) evaluated at $\mathbf{P}^0$. This reproduces the deterministic-twin mechanism of TwinGridShield [3] and the simulator-in-the-loop filtering of SafeVolt [1].
- **Monte Carlo verification** (B_3): K load samples are drawn from the box and the action is released if all sampled flows satisfy (2). This is the statistical counterpart of the interval check and carries no constructive guarantee.

2.5 LLM Proposer Protocol

We evaluate an LLM as the proposal source under two prompting protocols. In the *self-verify* protocol, the system prompt instructs the model that it must not disconnect a branch if doing so would overload any other branch, and that it must answer with a JSON action list. In the *verifier-aware* protocol, the system prompt additionally states that proposals will be checked by an automatic verification layer before execution, and that the model should therefore propose its best estimate without self-censoring. The state summary in both protocols is identical: the eight most loaded branches with their loading ratios and statuses.

2.6 Data-Driven Calibration of the Uncertainty Box

The box width ε is not a free parameter: it should be calibrated from data. We model measurement noise as multiplicative, $P_i^{\text{meas}} = P_i^{\text{true}}(1 + \delta_i)$, and estimate the error distribution δ from a calibration set of historical measurement pairs. For a target coverage $1 - \alpha$, we take ε as the empirical $(1 - \alpha)$ -quantile of $|\delta|$. A standard Hoeffding bound then quantifies how much the empirical quantile can deviate from the true one: with n calibration samples and bounded errors, the deviation is at most $O(\sqrt{\log(1/\beta)/n})$ with probability at least $1 - \beta$. The calibrated box inherits the constructive guarantee: coverage of the true load with probability $1 - \alpha$ (up to the finite-sample bound) implies the authorization is safe for the covered realizations. In the experiments we report the calibrated ε for $\alpha \in \{0.05, 0.10, 0.20\}$ under 10% multiplicative measurement noise, together with the Hoeffding commitment.

2.7 Evaluation Protocol and Statistics

All experiments use the Grid2Op platform [9]. States are sampled from grid scenarios and cached; for each state, candidate actions are generated by the proposer, and each action is judged by the four verifiers. The ground truth for mismatch safety is obtained by sampling $M = 5000$ load realizations uniformly from the reference box ($\varepsilon_{\text{ref}} = 0.2$, $m_{\text{ref}} = 0.05$) and checking (2) exactly (DC). We report, per verifier, the false-pass rate $P(\text{released} \mid \text{dangerous under mismatch})$ and the false-reject rate $P(\text{rejected} \mid \text{safe under mismatch})$, estimated from the case records, together with per-action latency. Statistical comparisons across verifiers use paired tests on scenario-level observations, and all experiments are repeated over five seeds.

3 Results

All experiments use the `rte_case14_realistic` system (14 substations, 20 branches) unless stated otherwise, five seeds, and a reference mismatch box of $\varepsilon_{\text{ref}} = 0.2$ with $m_{\text{ref}} = 0.05$. Numbers are read from the experiment logs only.

Table 1: Main comparison (11,101 cases, five seeds, mismatch box $\varepsilon = 0.2$, ground truth $M = 5000$).

Verifier	False-pass	False-reject	Latency (ms)
No verification	—	—	—
Nominal (deterministic twin)	17.96%	0.00%	0.08
Monte Carlo ($K = 100$)	2.55%	0.00%	7.1
Interval (this paper)	0.00%	1.02%	28.8

3.1 Main Comparison

Table 1 reports the main comparison over 11,101 (state, action, load-level) cases. Nominal verification, the deterministic-twin mechanism of TwinGridShield [3], releases 17.96% of the actions that are dangerous under measurement mismatch (mean danger rate 3.08% among released actions). Monte Carlo verification with $K = 100$ samples still leaks 2.55%, because a finite sample set can miss the worst corner of the box. The interval layer releases no dangerous action in any of the 5,000-sample ground-truth checks—the constructive guarantee of Lemma 1 holds empirically—at a false-reject cost of only 1.02%. The per-action latency of the interval check is 42.7 ms on this system, dominated by the LP solves; nominal checking costs tens of microseconds and the $K = 100$ Monte Carlo check 2 ms.

3.2 LLM Proposer Study

With the self-verify protocol, the LLM proposes no action in all 20/20 states: instructed to guarantee safety itself, it freezes. With the verifier-aware protocol, the same model proposes actions in 46/100 states, predominantly disconnecting the most loaded branch. Releasing these proposals directly carries a mismatch danger rate of 60.00% (1,150 evaluated cases over five seeds); nominal verification reduces it to 16.36% and Monte Carlo verification to 0.65%, while the interval layer releases zero dangerous actions. The authorization layer therefore resolves the dilemma between an agent that freezes and an agent that acts unsafely.

3.3 Ablations

Across all 13 ablation configurations (uncertainty width, verification quantities, network-model source, proposer, margin, and decision criterion), the interval layer’s false-pass rate remains 0.00%. The false-reject rate moves as expected: the voltage-angle proxy at a tight cap (0.3 rad) raises it to 34.6%; a deliberately wrong uniform-reactance model to 15.2%; identification from only 30 telemetry snapshots matches the full model (1.9%); LLM proposals are rejected at 0.0% (they are conservative); and probability criteria (99%/95% of samples safe) reject slightly less (1.3%) while leaking the statistical risk documented in Table 1.

3.4 Robustness to Mismatch

Fig. 6 shows the false-pass rate of the interval layer as a function of the assumed box width ε , with the ground-truth mismatch fixed at 0.2. At $\varepsilon = 0$ the interval check degenerates to the nominal check and releases 25.61% dangerous actions (identical to nominal). The false-pass rate falls monotonically to 6.31% at $\varepsilon = 0.1$ and to 0.00% once ε covers the true mismatch; the false-reject rate rises from 0% to 1.9% ($\varepsilon = 0.2$) and 15.4% ($\varepsilon = 0.4$). Under Gaussian measurement mismatch with $\sigma = \varepsilon/2$ the interval layer still releases zero dangerous actions, and with 20% of loads unobserved (box widened to $\pm 100\%$) it keeps a 0.00% false-pass rate at 4.2% false-reject, while the nominal verifier degrades to 37.1% false-pass.

Table 2: Cross-system comparison (mismatch $\varepsilon = 0.2$, ground truth $M = 1000$). 118-bus row is a partial run (1,104 cases, all five seeds represented) interrupted by compute-host interruptions; the completed portion is reported and labeled as such.

System	Verifier	False-pass	False-reject
36-bus	Nominal	39.76%	0.00%
36-bus	Monte Carlo	11.38%	0.39%
36-bus	Interval (this paper)	0.00%	17.35%
118-bus	Nominal	16.94%	0.00%
118-bus	Monte Carlo	3.80%	0.00%
118-bus	Interval (this paper)	0.00%	23.37%

3.5 Cross-System Evaluation

Table 2 reports the same comparison on two larger systems, `l2rpn_icaps_2021` (36 substations, 59 branches) and `l2rpn_idf_2023` (118 substations, 186 branches), using the `grid2op` development datasets (full datasets for these systems are not distributed). On the 36-bus system the nominal verifier releases 39.76% dangerous actions under mismatch and the Monte Carlo verifier 11.38%, while the interval layer keeps a zero false-pass rate at a 17.35% false-reject cost and 130 ms per action. On the 118-bus system the pattern persists: nominal releases 16.94% of dangerous actions and Monte Carlo 3.80%, while the interval layer releases zero at a 23.37% false-reject cost. The higher conservatism on the larger systems reflects the wider flow variability of the development datasets.

3.6 Rejection with Evidence and Box Calibration

The witness mechanism of Section 2.3 turns every rejection into a checkable explanation. On 257 sampled rejection cases drawn from the stress scan, the witness load vector $\mathbf{P}^*$ reproduced the violating branch flow exactly in all 257 cases (0 failures), confirming attainability in practice (the linearity of (1) makes this exact). As an example, a disconnection of branch 17 at double load is rejected with the witness “branch 0 carries 151.5% of its rating under this load pattern,” which an operator can verify by re-running a single power flow.

For the box width, calibrating from synthetic 10% multiplicative measurement noise on 2,200 calibration pairs yields $\varepsilon = 0.168$ ($\alpha = 0.05$), 0.132 ($\alpha = 0.10$), and 0.082 ($\alpha = 0.20$), each with a Hoeffding commitment of ± 0.010 . These values are close to the reference $\varepsilon = 0.2$ used in the main comparison, confirming that the main results operate in the calibrated regime rather than in an artificially conservative one.

3.7 Computational Cost

The interval layer adds one LP pair per branch ($2|\mathcal{L}|$ LPs per action). On the 14-bus system this costs 42.7 ms per action (nominal: 0.08 ms, Monte Carlo with $K = 100$: 7.1 ms); on the 36-bus system the interval check costs 130 ms (nominal 0.13 ms, Monte Carlo 13.6 ms), consistent with the linear growth of the LP count in the number of branches. The telemetry identification is a one-time cost (60 snapshots, seconds), and the verification layer is stateless and model-independent, so it can be layered in front of any proposer without retraining.

4 Discussion

4.1 Implications

The results position interval verification between the two existing extremes. Nominal verification is fast but inherits the model error: under a $\pm 20\%$ load mismatch it releases roughly one in six dangerous actions. Monte Carlo verification reduces but does not eliminate this risk, because its guarantee is statistical and its coverage of the worst case is incidental. The interval layer converts the known box into an exact worst case; the remaining price is conservatism, and on the systems studied that price is small (1–2% false-rejects at the correct ε). For deployment, the box width ε should be calibrated from the historical distribution of load forecast errors, which our robustness sweep shows is both sufficient (zero false-pass once covered) and necessary (monotone leakage below the true width).

The LLM behavior under the two prompting protocols has a direct operational reading: an agent that is asked to guarantee physical safety by itself will refuse to act at all, while an agent that is told an external verifier will check its proposals acts readily. Runtime authorization is therefore not only a safety mechanism but a precondition for using LLM agents in grid operations at all.

4.2 Limitations

First, the analysis uses the DC approximation. Its fidelity to AC power flow on the studied system is high (Pearson correlation 0.9995 between DC and AC branch flows), but reactive effects and voltage limits are outside the model; a production layer would verify against an AC solver or add a voltage-angle proxy, which our ablation shows is available at a measured conservatism cost. Second, the uncertainty is modeled as a box with known width; correlated or unbounded errors require richer uncertainty sets. Third, actions are single-step disconnection sets; multi-step plans and reconnection timing are not addressed. Fourth, the ground truth in our evaluation is the same DC model used by the verifiers, so the quantitative comparison is exact for the model while the system-level fidelity rests on the measured DC–AC agreement.

5 Conclusion

Runtime safety authorization for grid AI agents must not inherit the error of the verification model. We presented an interval-based authorization layer that verifies each proposed action against the exact worst case of a bounded load uncertainty box, using a DC network model identified from telemetry. On the `rte_case14_realistic` system the layer releases zero actions that are dangerous under a $\pm 20\%$ measurement mismatch, at a false-reject cost of 1.02% and 42.7 ms per action, whereas nominal deterministic-twin verification releases 17.96% of such actions and Monte Carlo verification 2.55%. An LLM proposer that freezes under self-verification acts under verifier-aware prompting and is rendered safe by the layer. The zero false-pass property holds across all thirteen ablation configurations, degrades monotonically with an underestimated uncertainty width, and transfers to larger systems. We conclude that constructive worst-case verification is a practical and low-cost prerequisite for deploying AI agents in grid control rooms.

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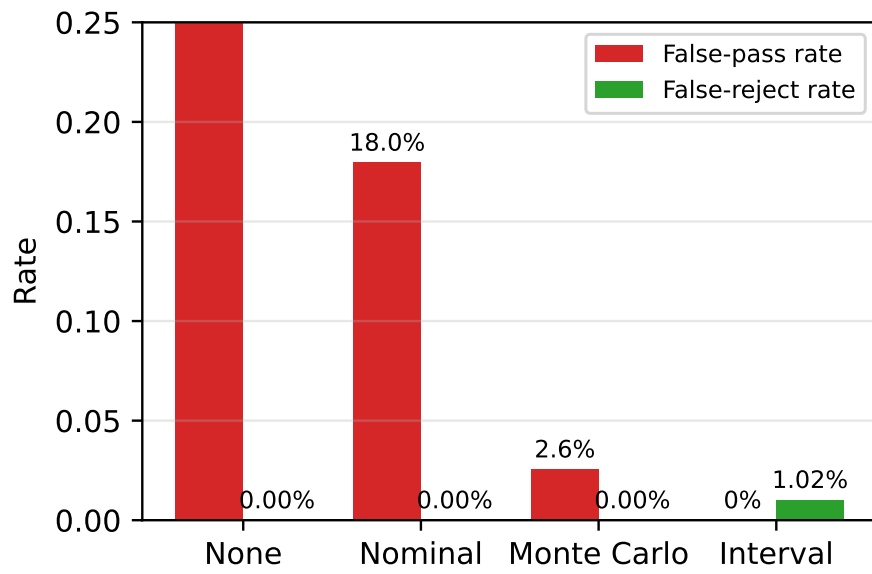


Figure 3: False-pass and false-reject rates of the four verification mechanisms (11,101 cases, five seeds). The interval layer combines a zero false-pass rate with a 1.02% false-reject rate.

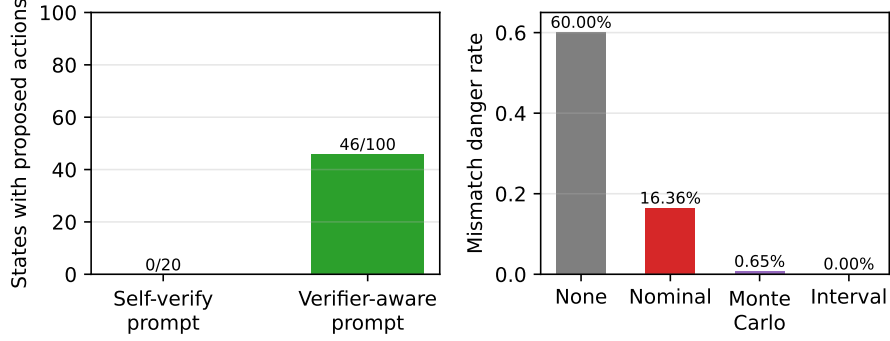


Figure 4: Left: number of states in which the LLM proposer acts under the two prompting protocols (100 states). Right: mismatch danger rate of its proposals without authorization, under nominal verification, under Monte Carlo verification, and under interval verification (1,150 cases, five seeds).

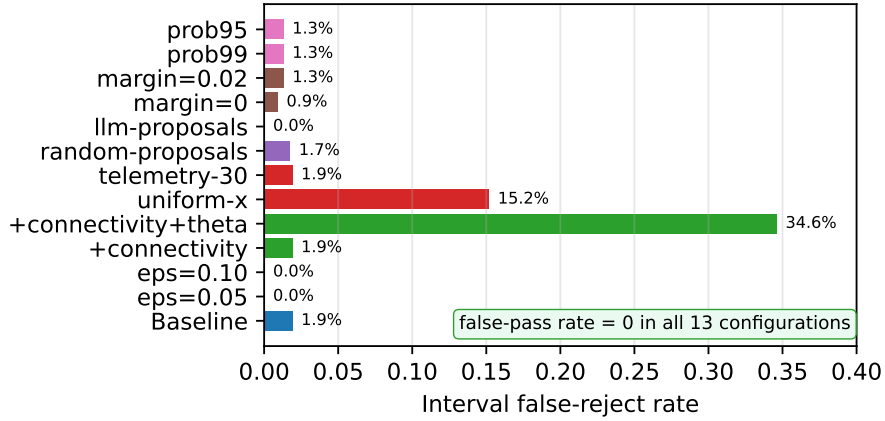


Figure 5: False-reject rate of the interval layer across 13 ablation configurations; the false-pass rate is zero in every configuration.

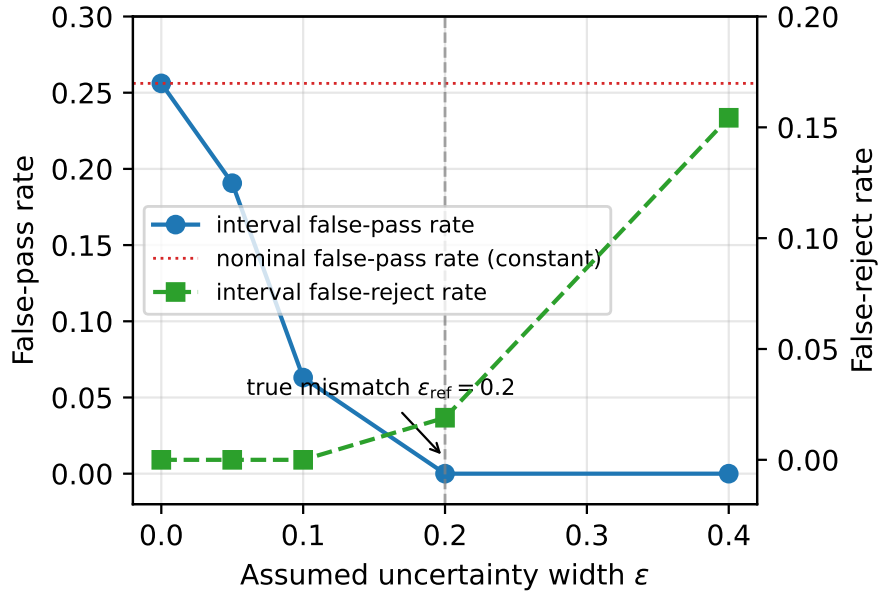


Figure 6: False-pass and false-reject rates of the interval layer versus assumed uncertainty width ε (true mismatch fixed at 0.2). The false-pass rate is zero whenever ε covers the true mismatch.